

OSMAN BUGRA DAGLI

Milano, MI (Open to relocate) – +39 3481899303 – osman.dagli687@gmail.com – github.com/osmandagli

EDUCATION

Politecnico di Milano

Master of Engineering

Cumulative GPA: 24.43/30.00

Advanced OS, IoT

Ankara University

Bachelor of Engineering / Computer Engineering

Cumulative GPA: 3.81/4.0

Milano, MI

Sep 2024 – December 2026 (Expected)

Ankara, ANK

Sep 2018 – June 2023

EXPERIENCE

TUSAS (Turkish Aerospace Industries)

DevOps Engineer

Ankara, Turkey

Nov 2022 – Sep 2024

- Built and maintained containerized infrastructures for Kubernetes to streamline operations.
- Led the migration of application servers (e.g., Tomcat) from traditional OS-based deployments to Kubernetes.
- Troubleshoot and optimized GitLab CI/CD pipelines to ensure seamless integration and deployment.
- Developed automation scripts (Bash, Python) to reduce workload and increase productivity.

ASELSAN Electronic Industries Inc.

DevOps Intern

Ankara, Turkey

Aug 2022 – Sep 2022

- Implemented legacy software that can't be containerized on Kubernetes using KubeVirt.

THESIS

Accelerating a Media-over-QUIC (MoQ) Relay with AF_XDP Kernel-Bypass Networking

Master's Thesis

Politecnico di Milano

Jan 2025 – Present

- Profiled the data-path CPU cost of Meta's moxygen MoQ relay, building a perf-based CPU-time breakdown pipeline that attributed a significant share of per-core time to kernel packet I/O and identified the relay's design.
- Proved an AF_XDP zero-copy fast path sustaining ~1.55 Mpps single-core RX vs. ~0.65 Mpps for the tuned kernel recvmmsg+GRO baseline, where a ~2.4 per-core gain.
- Designed a non-invasive socket-injection seam into the QUIC stack that redirects the relay's entire UDP RX/TX path through an AF_XDP socket with zero source changes.

PROJECTS

Minimal container runtime in Go

March 2026

<https://github.com/osmandagli/containerrrt>

- Built a container runtime from scratch on raw Linux primitives
- Implemented process isolation with UTS, PID, and mount namespaces, and filesystem isolation via pivot_root (fully unmounting the host root rather than hiding it as chroot does), plus an ephemeral /dev tmpfs with the essential device nodes.
- Applied cgroups v2 resource limits using a pipe-based synchronization barrier between parent and child, guaranteeing the workload starts already inside its limits and cannot race the setup.

Connection Tracker

May 2025

<https://github.com/osmandagli/conntracker>

- Developed ConnTracker, written in eBPF/XDP, used for connection tracking & fingerprinting system.
- Built kernel-level logic and TCP and UDP state machines to monitor connection metrics and extract TCP fingerprint data.
- Used probabilistic data structures (Count-Min sketch, Bloom filters) for efficient and fast threshold-triggered filtering.

ADDITIONAL

Technical Skills

Advanced in Go, Python, GitLab CI/CD, Helm; Proficient in Kubernetes, Bash, Linux

Languages

English (Professional), Italian (B1), Turkish (Native)

Certifications

CKA (Certified Kubernetes Administrator)

Activities

Table tennis team player, AON Milano Sport Tennistavolo (Italian league, Sep 2024–Present)